

ST JAMES, MN, USA

WMO: 726554

Lat: 43.986N

Lon: 94.558W

Elev: 1067

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54906

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.4	-6.1	-17.3	2.2	-9.4	-13.0	2.8	-5.4	32.7	19.2	29.3	16.0	8.5	300	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	90.2	74.4	87.9	73.0	84.2	71.4	79.2	84.7	76.8	83.8	74.7	81.4	12.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	148.3	82.9	74.7	135.1	81.2	72.5	125.2	79.1	43.7	83.9	41.0	84.0	39.0	82.1	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.9	24.9	21.9	-16.7	95.2	5.2	2.9	-20.4	97.2	-23.4	98.9	-26.3	100.5	-30.1	102.6		
			-17.1	82.1	5.1	2.9	-20.8	84.2	-23.8	85.9	-26.6	87.5	-30.3	89.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	17.1	21.0	32.8	46.8	59.4	68.9	72.3	69.7	63.1	49.7	35.6	21.6
	DBStd	21.95	12.77	12.51	12.35	10.23	9.00	6.75	5.48	5.55	8.42	9.58	11.06	12.09
	HDD50	4031	1019	813	547	175	20	0	0	0	7	124	445	881
	HDD65	7395	1484	1233	998	550	222	38	8	19	134	482	881	1346
	CDD50	2800	0	0	14	79	311	566	692	610	399	115	14	0
	CDD65	687	0	0	0	3	47	154	235	163	77	8	0	0
	CDH74	6414	0	0	3	68	566	1403	2192	1313	751	114	4	0
	CDH80	2006	0	0	0	17	210	473	705	346	227	28	0	0
(14) Wind	WSAvg	9.3	10.8	10.7	10.4	10.8	10.8	8.6	6.2	5.6	7.3	9.3	10.4	10.4
(15) Precipitation	PrecAvg	30.3	0.7	0.7	1.7	3.2	4.1	4.8	3.9	3.7	3.0	2.2	1.3	0.9
	PrecMax	45.3	1.5	1.8	3.1	6.5	6.8	10.2	7.7	9.8	8.7	6.5	4.1	2.4
	PrecMin	20.3	0.1	0.0	0.1	1.0	1.3	2.3	0.7	0.7	0.4	0.3	0.1	0.1
	PrecStd	6.1	0.4	0.4	0.9	1.4	1.6	1.9	1.8	2.0	2.3	1.6	1.0	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	54.7	72.0	81.9	91.3	93.0	91.5	90.3	90.3	83.8	70.4	54.2
		MCWB	40.3	48.6	57.1	61.1	67.2	72.1	78.3	78.3	71.4	63.2	55.3	49.3
	2%	DB	39.3	46.4	63.2	73.4	85.8	89.6	90.1	87.6	85.9	76.6	63.4	45.9
		MCWB	35.7	42.6	54.0	55.8	64.3	71.3	77.7	76.9	70.3	60.1	52.9	40.7
	5%	DB	36.6	40.8	55.5	69.8	81.0	84.5	86.5	83.7	82.0	71.5	56.9	39.5
		MCWB	34.2	37.3	49.4	54.5	62.5	69.0	75.7	74.3	67.9	57.8	48.3	36.3
	10%	DB	34.2	36.7	49.9	63.7	74.9	81.8	83.8	81.3	78.9	64.5	52.1	36.7
		MCWB	32.6	34.5	45.6	52.4	59.9	68.0	74.1	72.2	65.8	55.2	45.6	34.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.3	49.8	61.1	64.8	73.7	79.0	83.6	82.4	76.3	70.8	59.1	49.4
		MCDB	47.5	54.0	65.7	74.7	83.4	85.0	90.0	86.1	84.6	77.1	66.6	53.3
	2%	WB	36.7	42.7	55.0	60.3	68.2	75.5	79.8	79.0	73.4	63.4	54.0	41.2
		MCDB	39.3	46.6	60.9	70.0	79.5	84.4	85.7	83.6	81.3	69.1	61.8	45.7
	5%	WB	34.4	37.8	50.4	56.7	65.5	72.9	77.4	75.9	71.0	60.1	49.8	37.4
		MCDB	36.4	40.2	55.6	65.8	75.6	80.5	84.7	81.6	79.3	68.1	56.6	39.9
	10%	WB	32.3	34.8	45.3	53.4	62.7	70.6	75.4	73.4	68.2	56.6	45.5	34.7
		MCDB	34.0	36.8	48.9	62.1	71.7	78.6	82.0	79.0	75.5	64.8	49.9	36.8
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	16.0	16.1	19.9	20.5	19.0	19.5	19.6	22.7	21.0	17.8	14.9
		MCDBR	17.4	19.2	25.1	30.0	28.0	23.3	21.0	21.4	26.1	28.9	26.6	19.6
		MCWBR	14.5	15.4	17.2	16.2	12.5	10.6	12.0	12.6	12.6	14.8	16.4	15.6
	5% WB	MCDBR	16.3	17.8	23.3	24.8	22.5	19.5	19.0	18.8	22.3	22.4	24.5	18.4
		MCWBR	13.7	14.4	16.9	15.5	12.1	11.6	12.4	12.8	12.6	14.2	16.5	14.9
(40) Clear-Sky Solar Irradiance	taub		0.266	0.284	0.310	0.363	0.383	0.403	0.415	0.411	0.369	0.325	0.285	0.264
	taud		2.453	2.406	2.451	2.337	2.347	2.341	2.293	2.286	2.376	2.496	2.528	2.474
	Ebn at Noon		279	291	296	285	282	275	270	267	271	272	269	268
	Edh at Noon		23	28	31	38	39	39	41	39	33	26	21	20
(44) All-Sky Solar Radiation	RadAvg		580	911	1198	1476	1725	1915	2058	1704	1359	895	589	437
	RadStd		64	105	121	144	128	150	81	126	94	117	51	42

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page